

Proof: As mentioned in Lemma 1, the topology of weak convergence on $\mathcal{P}$ is metrizable with the Lévy-Prohorov metric [15] and defined as

$$L(\mu, \nu) = \inf \left\{ \delta : \mu(A) \leq \nu(A^{(\delta)}) + \delta \text{ and } \nu(A) \leq \mu(A^{(\delta)}) + \delta \text{ for all } A \subseteq \mathcal{B} \right\} \quad (125)$$

for any $\mu, \nu \in \mathcal{P}$, where $A^{(\delta)}$ denotes the set of all $x \in \mathbb{R}$ which lie a d -distance less than δ from A .

Similarly as in the proof of Lemma 1, it suffices to show that $\Lambda_0(\gamma, q_0)$ is both tight and closed in $\mathcal{P}$. The tightness can be shown by the same arguments as in Lemma 1. In the following, we prove that $\Lambda_0(\gamma, q_0)$ is closed in $\mathcal{P}$.

Let $B_m = [-\frac{1}{m}, \frac{1}{m}]$ for $m = 1, 2, \dots$. Let $\{\mu_n\}_{n=1}^\infty$ be a convergent sequence in $\Lambda_0(\gamma, q_0)$ with limit μ_0 . For any $m \in \mathbb{N}$, there exists an n_m such that $L(\mu_n, \mu_0) < \frac{1}{m}$ for all $n > n_m$. By the definition of L in (125), we have for any $m \in \mathbb{N}$ and $n > n_m$,

$$\mu_0(\{0\}) \leq \mu_n(B_m) + \frac{1}{m}, \quad (126)$$

and

$$\mu_n(\{0\}) \leq \mu_0(B_m) + \frac{1}{m}. \quad (127)$$

For any $n \in \mathbb{N} \cup \{0\}$, we have

$$\mu_n(\{0\}) = \mu_n \left(\bigcap_{m=1}^\infty B_m \right) = \lim_{m \rightarrow \infty} \mu_n(B_m), \quad (128)$$

so for any $m \in \mathbb{N}$, there exists an n'_m such that $\mu_n(B_m) \leq \mu_n(\{0\}) + \frac{1}{m}$. Therefore, according to (126) and (127), for all $m \in \mathbb{N}$ and $n > \max\{n_m, n'_m\}$,

$$q_0 - \frac{2}{m} \leq \mu_0(\{0\}) \leq q_0 + \frac{2}{m}. \quad (129)$$

Thus we have $\mu_0(\{0\}) = q_0$ by letting $m \rightarrow \infty$.

Moreover, let $f(x) = x^2$ which is continuous and bounded below. By weak convergence [15, Section 3.1], we have

$$\mathbb{E}_{\mu_0} \{X^2\} = \int f d\mu_0 \leq \liminf_{n \rightarrow \infty} \int f d\mu_n \leq \gamma. \quad (130)$$